6 (a) (i) the spreading of the waves when it passes through a gap or around an obstacle.
B1

Comments:

This part is generally not well done by most students. Most of them did not use the key word " spreading " when describing the phenomenon, and even if they do, they did not state that the phenomena occurs when passing through a gap or an obstacle.

(ii) wavelength of the electron must be comparable/of the same order of magnitude as the atomic spacing of the graphite. B1

Comments:
This part is badly done by most students. Most of them were not able to state the condition for diffraction to occur which is rather alarming.

(b) (i) Wavelength associated with a particle that is moving. B1

Comments:
This part is also not well done. Most of the students just either quote the equation for de_broglie wavelength without explaining what it is, or give the wrong definition.

(ii) work done by p.d = increase in ke of electron B1

$$qV = \frac{1}{2}mv^2 = \frac{p^2}{2m} \quad \text{since } p = mv$$ B1

Hence, $p = \sqrt{2mqV}$

$$p = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 120} = 5.91 \times 10^{-24} \text{ kg m s}^{-1}$$ C1

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{5.91 \times 10^{-24}} = 1.12 \times 10^{-10} \text{ m}$$ A1

Comments:
Most students were able to obtain the correct answer. Those weaker students use the speed of light as the speed of the particle which is a misconception.

(iii) When the p.d $V$ is increased slightly, the momentum of the electron would increase. Hence, the de Broglie wavelength of the electron would decrease.
B1

Using $d \sin\theta = n \lambda$, the angular separation between successive fringes would decrease,
C1

and therefore diameter of the concentric circles decreases. A1

Comments:

This part is generally not very well done. Some of the weaker students use the young double slit formula in explaining which is not correct, instead the correct formula is the diffraction grating formula.